

As a result of the restrictions on the permissible couplings in the prototype, the even- and odd-mode equivalent circuits do not contain bridge couplings. Hence the input admittance to the even- or odd-mode circuit can be developed as a continued fraction:

$$y_{\pm} = j(B_1 \pm K_{1,n+2}) + \frac{(K_{12} \pm K_{1,n+1})^2}{s + j(B_2 \pm K_{2,n+1}) + \frac{(K_{23} \pm K_{2,n})^2}{s + \dots}} \quad (1)$$

where $+$ ($-$) refers to the even (odd) mode. When n is odd, the degree of y_{-} is one less than the degree of y_{+} because the short circuit shunts the centre resonator in the odd mode. The open-circuit impedance parameters of the prototype are given by the well known relationships $z_{11} = z_{22} = (z_{+} + z_{-})/2$ and $z_{12} = z_{21} = (z_{+} - z_{-})/2$, where $z_{\pm} = 1/y_{\pm}$.

Synthesis: Schoeffler⁶ and Bown⁷ have described similar synthesis procedures for symmetric lattices that may contain constant reactances, and both procedures are directly applicable to the coupled-resonator prototype. Bown's method is more general, and is preferable from a numerical standpoint because it leads directly to immittances that do not contain surplus factors. The following procedure is equivalent to Bown's:

(a) Choose a characteristic function $\phi|\omega|$ such that $|\phi|^2 = (P/Q)^2$, where P and Q are real polynomials in ω .

(b) Factor $P + jQ = (P_1 + jQ_1)(P_2 - jQ_2)$, where $P_1 + jQ_1$ contains all the upper-half ω plane zeros and P_1 , Q_1 , P_2 and Q_2 are real polynomials. Thus $P_1 + jQ_1$ contains some of the natural frequencies, and $P_2 + jQ_2$ contains the remaining natural frequencies.

(c) Form the polynomials $P_i' + jQ_i' = e^{j\theta}(P_i + jQ_i)$, $i = 1, 2$. The P_i' and Q_i' are real polynomials, and θ is an arbitrary phase angle.

(d) The normalised lattice-arm immittances are

$$w_i = \frac{jQ_i'}{P_i'} \quad i = 1, 2 \quad (2)$$

These immittances can be equated to y_{+} and y_{-} with the following restrictions:

(i) As required by eqn. 1, a numerator of w_i cannot be of higher degree in ω than the denominator. If necessary, this can be corrected by a different choice of θ ; e.g. shifting θ by $\pm 90^\circ$ will invert the immittances.

(ii) When n is odd, the lower-degree immittance must be assigned to y_{-} .

The comparison of the expansions of y_{+} and y_{-} in eqn. 1 on a term-by-term basis will yield the complete set of element values for the filter, thus completing the synthesis.

Discussion: The choice of real polynomials $P(\omega)$ and $Q(\omega)$ in step (a) above is necessary and sufficient for $|\phi|^2$ to be realisable by a symmetric, lossless doubly-terminated LCX network.⁶ Thus P , whose zeros are the loss zeros, and Q , whose zeros are the finite-frequency loss poles, are not required to be even or odd functions of frequency. All lowpass responses of this type can be realised by the prototype of Fig. 2 through the use of eqn. 2, including general responses that are not symmetric about $\omega = 0$, responses that have finite loss at $\omega = \infty$, such as even-degree elliptic-function responses, and nonminimum phase responses. Further, the expansion of y_{+} and y_{-} into the continued fraction of eqn. 1, which is similar to a canonical Cauer form, guarantees that the number of couplings and resonators in the network will be the minimum necessary to realise the desired z parameters. For these reasons, the coupled-resonator prototype presented here is a canonical realisation.

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LOW-FREQUENCY NOISE IN GaAs SCHOTTKY-GATE F.E.T.S

Indexing terms: Field-effect transistors, Noise

Measurements of the low-frequency noise in GaAs Schottky-gate f.e.t. channels have been performed from 10 Hz to 0.5 MHz between -120 and $+21^\circ\text{C}$. Diffusion and recombination noise sources appear to be predominant at low temperatures for low-power-level conditions.

Nous avons entrepris la mesure du bruit basse fréquence prenant naissance dans le canal des f.e.t. à l'arséniure de gallium pour des fréquences allant de 10 Hz à 0.5 MHz et une température comprise entre -120 et $+21^\circ\text{C}$. Nous avons mis en évidence que pour de faibles puissances dissipées les sources de bruit principales à basse température ont pour origine des phénomènes de diffusion et de génération recombinaison.

High-frequency-noise behaviour of GaAs Schottky-barrier field-effect transistors (f.e.t.) cannot be fully explained by thermal noise in the channel or in the contact resistors, nor can it be fully explained by the induced gate noise source, even by using the hot-carrier hypothesis. Besides, the noise contribution due to intervalley scattering¹ seems quite irrelevant.² Among numerous hypotheses, the influence of low-frequency-noise sources has never been taken into account, although it has been shown that low-frequency-noise sources may have a real influence on the high-frequency-noise performance of microwave devices.³ Further, low-frequency-noise measurements are a tool to characterise the defects and traps in semiconductors.⁴⁻⁶ To obtain a better understanding of this noise, we have carried out measurements on the epitaxial layer of a GaAs f.e.t.*, regarding these devices as single dipoles having their source and gate earthed (Fig. 1). When the drain voltage is far below pinch-off, the space-charge region is much narrower than the total epitaxial-layer thickness. Thus it can be assumed that the gate influence is quite negligible.

Noise variation with current: According to our previous hypothesis, the device is biased in the ohmic region of the I_D/V_{DS} curve and the voltage noise in the channel increases linearly with the $I_D V_{DS}$ product (equivalent to V_{DS}^2). This result is characteristic of noise due to carrier fluctuation such as generation-recombination (g.r.), flicker or diffusion noises, and bear out that, for low-level powers, carrier fluctuations

* Plessey GAT 1

in the gate space-charge region have no influence on noise. Otherwise, the noise amplitude against V_{DS} curve would have shown, on logarithmic scales, a slope of between 2 and 5/2, by taking into account the modulation of the space-charge-region width.⁷

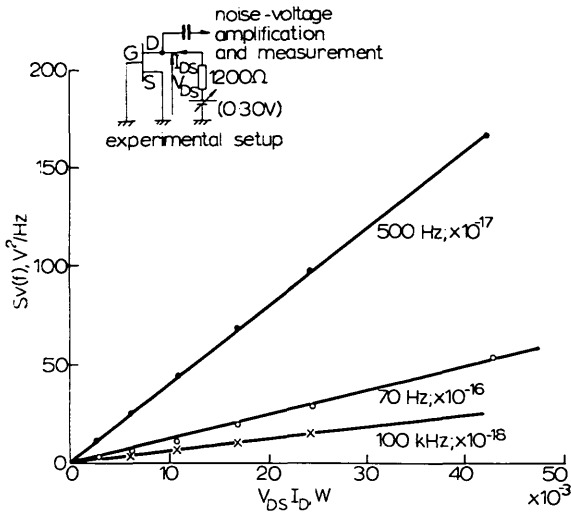


Fig. 1 Experimental setup for low-frequency-noise measurements and spectral voltage-density variations against dissipated power in channel

Device number 16; $T = -92^\circ\text{C}$

Noise spectra (10 Hz to 0.5 MHz): In Fig. 2 are shown (broken lines) three different components of the noise spectrum at a temperature of -92°C . The first, with an f^{-1} slope, makes evident the presence of flicker noise. The second, 100 to 10 000 Hz, is a transition region corresponding to a 'plateau', possibly due to a generation-recombination process. The third, 10 000 Hz to 0.5 MHz, presents an $f^{-3/2}$ slope, which can be interpreted as:

- (a) a generation-recombination process with a time-constant distribution
- (b) a single-time-constant generation-recombination process modified by diffusion noise.

In the first case, the noise spectrum would have been of the form $f^{-\alpha}$, with $1 < \alpha < 2$, but not necessarily equal to 1.5. Up to now, we have never obtained spectra where α differs from 1.5. So it appears that such hypothesis is no longer necessary. In the second case, Van Vliet calculations⁸ lead to a spectrum of the form $f^{-1.5}$. Indeed, the spectral density of carrier fluctuations can be expressed by

$$G\Delta n(\omega) = 4(\Delta n^2) \left\{ \frac{\tau}{1 + \omega^2 \tau^2} + \frac{1}{L} \left(\frac{D}{2\omega^3} \right)^{\frac{1}{2}} \psi(\theta) \right\} \quad (1)$$

with

- n = carrier density
- τ = time constant
- L = device length
- D = diffusion coefficient

and

$$\psi(\theta) = \frac{\theta^{5/2} \{ (1 + \theta^2)^{1/2} - 1 \}^{1/2} - \theta^{3/2} \{ (1 + \theta^2)^{1/2} + 1 \}^{1/2}}{(1 + \theta^2)^{3/2}} \quad (2)$$

where $\theta = \omega\tau$. For a bulk semiconductor with N_t traps, biased with a constant current I_{DS} , the Δn^2 coefficient can be calculated.⁹ Thus, taking into account $1/f$ noise, the total spectral voltage density is

$$S_v(f) = \frac{C}{f} + \frac{4V_{DS}^2}{n^2 V} \frac{N_0(N_t - N_0)}{N_t} \left\{ \frac{1/2\pi f_0}{1 + (f/f_0)^2} + \frac{\alpha \psi(\theta)}{f^{3/2}} \right\} \quad (3)$$

where C is the flicker-noise coefficient, V_{DS} is the drain-source voltage, equal to RI_{DS} if $1/R$ is the constant slope in the ohmic region of the I_D/V_{DS} characteristic, N_0 is the density of occupied traps, $f_0 = 1/2\pi\tau$ and the α coefficient is related to the length of the device, the channel in this case, by

$$\alpha = \frac{1}{L} \left(\frac{D}{2} \right)^{1/2} \left(\frac{1}{2\pi} \right)^{3/2} \quad (4)$$

Noise variation with temperature: Fig. 2 shows the variation of noise amplitude against temperature between -120 and $+21^\circ\text{C}$. It is verified that flicker noise is not very sensitive to temperature. It appears that g.r. and diffusion noise are predominant in the low-temperature range (less than -50°C). It would be very useful to separate these three components. Unfortunately, such work is very hard to perform because of the large number of parameters and of the relative inaccuracy (10%) of noise measurements. However, in some cases, we can perform it for particular temperature conditions. Thus, by calculating 'plateau' values and time-constant evolution, we can obtain information about traps or defects responsible for noise.^{4, 9}

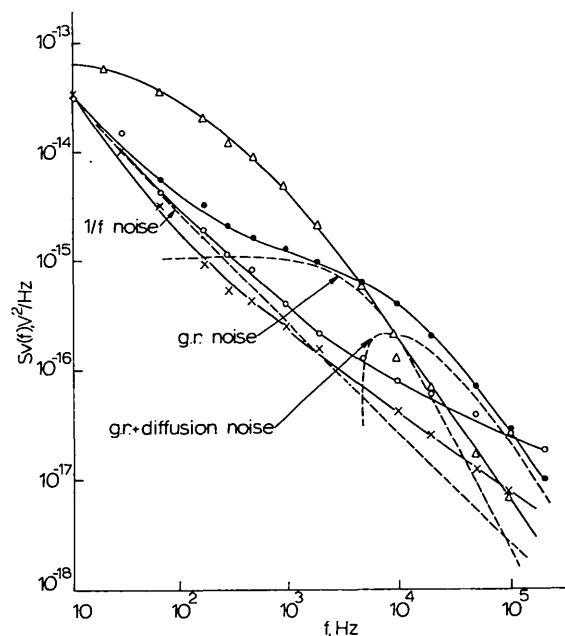


Fig. 2 Spectral voltage-density variations against frequency for different temperatures

$V_{DS} = 400$ mV
Broken lines represent the partition into three components, g.r., diffusion and flicker noise, of the -92°C curve
△ -120°C
● -92°C (---)
× -59°C
○ 21°C

Conclusion: Thus, in spite of a relatively large amount error, we have the important results:

- (a) The variation of the Df_0 product with temperature is very weak. Such a result is characteristic of a trap near the conduction band in this case (less than 0.1 eV).^{2, 3}
- (b) The diffusion coefficients computed from eqn. 4 are between ten and 100 times smaller than those calculated from the Einstein relation, taking a T^{-1} mobility-variation law with temperature T . This discrepancy can be explained from the fact that such a semiconductor with a carrier concentration of $4.5 \times 10^{16} \text{ cm}^{-3}$ at an ambient temperature degenerates at low temperatures, and, in consequence, the Einstein relation no longer holds.¹⁰

In conclusion, it appears that, in an GaAs f.e.t. channel with practically no space-charge region, flicker noise, which is the main low-frequency-noise source at ambient temperatures, is overtaken at lower temperatures by generation-recombination and diffusion sources. The behaviour of noise in a GaAs

Schottky-gate f.e.t. normally biased will now be studied. Thus the gate influence on noise will have to await the experimental evidence.

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IMPROVED ALGORITHM FOR STATE-SPACE SYNTHESIS

Indexing terms: Network-synthesis, State-space methods

Given a positive real $Z(s)$ having a realisation $[A, B, C]$ an improved algorithm for determining a realisation $[A, B, L]$ of $W(s)$ required in the well known state-space-synthesis procedures is proposed.

The problem of realisation from an impedance matrix has been widely investigated in recent years. A triple $[A, B, C]$ of real constant matrices is to be a realisation of $Z(s)$ if $Z(s) = C(sI - A)^{-1}B$. The well known state-space realisation procedures^{1,2} require the determination of $W(s)$ such that $Z(s) + Z'(-s) = W'(-s)W(s)$, where $Z(s)$ is the given positive real matrix to be synthesised. This spectral factorisation³ is the first step for determining a symmetric positive-definite matrix P that is subsequently required in the realisation techniques. It has been shown in Reference 1 that, if $Z(s)$ has a realisation $[A, B, C]$, $W(s)$ has a realisation $[A, B, L]$, which is to be determined to find the symmetric positive-definite matrix P from the well known relation

$$PA + A'P = -L'L$$

For $Z(s)$ of order 1×1 , $[A, B, L]$ can be easily determined, but, for $Z(s)$ of higher order, the authors feel that there is no specific published procedure for finding the realisation of $W(s)$ such that $[A, B]$ of $Z(s)$ and that of $W(s)$ are same.⁴ The purpose of this letter is to present one such procedure.

Given $Z(s)$, or $W(s)$ obtained by spectral factorisation, there exist several algorithms for finding minimal realisations.

However, the technique of Ho and Kalman⁵ based on determining Markov parameters⁵ ($CB, CAB, \dots, CA^{n-1}B$) etc. has been acclaimed to be the simplest, especially for when $Z(s)$ or $W(s)$ are of order greater than 1. Hence, in general, it will be worth while to determine their realisation by this technique. It is interesting to note that, whereas the determination of Markov parameters helps in making the realisation technique simpler, it also serves as a prerequisite in finding the realisation of $W(s)$ such that $[A, B]$ of $H(s)$ and that of $W(s)$ are the same. This is illustrated as a result in the form of the following lemma, which can be easily verified:

Lemma: If the Markov parameters $[CB, CAB, \dots, CA^{n-1}B]$ and the sequence $[B, AB, \dots, A^{n-1}B]$ of a minimal realisation are given, C can be determined by

$$C = [CB, CAB, \dots, CA^{n-1}B][B, AB, \dots, A^{n-1}B]^{-1}$$

where $[\]^{-1}$ denotes the pseudo-inverse.

Hence the synthesis procedure of Anderson and Newcomb¹ can be modified as follows:

Given a positive real $Z(s)$,

(a) Determine the Markov parameters $[CB, CAB, \dots, CA^{n-1}B]$ of $Z(s)$, which can be easily determined by dividing each entry of the numerator of $Z(s)$ by the common denominator. Obtain the realisation $[A, B, C]$ by the technique given by Ho and Kalman.⁵

(b) Determine $W(s)$ by spectral factorisation³ and obtain the sequence of Markov parameters $[LB, LAB, \dots, LA^{n-1}B]$ of $W(s)$.

(c) B, AB etc. of $W(s)$ are known, these being the same as those of $Z(s)$, determined in step (a); therefore determine L by applying the above lemma.

Well known techniques exist for the rest of the synthesis procedure.^{1,2}

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ERRATA

MAJITHIA, J. C., and BHAR, R.: 'Optimum packet size in a packet-switched data network', *Electron. Lett.*, 1974, 10, pp. 191-193.

The authors would like to make the following corrections to their letter:

In eqn. 1, F_n should be F_h

Eqn. 4 should read

$$F_h = \left(1 + \frac{0.5P}{\mu + 0.5P} \right) \frac{K}{P}$$